

NPTEL

Stable Diffusion
Based on Latent Diffusion Model

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Motivation

Problem with Pixel-Space Diffusion

DDPM performs forward and reverse diffusion directly on high-dimensional pixel-space images.

$$x \in \mathbb{R}^{H \times W \times 3}$$

Example:

- ✓ A $512 \times 512 \times 3$ RGB image contains 786,432 pixel values.
- ✓ These must be processed during every denoising step, making diffusion computationally expensive.

Problems:

- ✓ High memory consumption
- ✓ High computational cost
- ✓ Slow inference due to hundreds of denoising steps

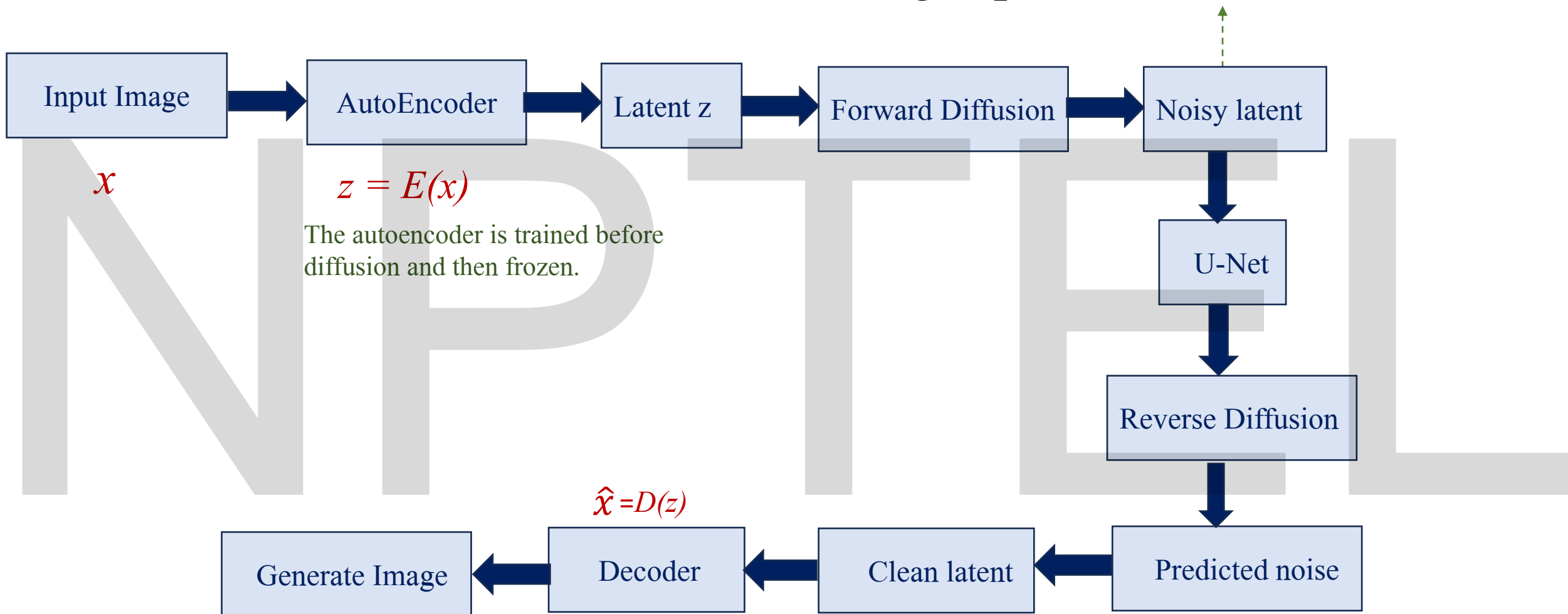
Latent Diffusion

Core Idea

- Compress the image into a compact latent representation and perform diffusion in latent space.
 - Introduced by R. Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," *Conference on Computer Vision and Pattern Recognition (CVPR)* (2022)
- ✓ Latent Diffusion Models (LDMs) are the algorithm introduced in the paper.
 - ✓ Stable Diffusion is the practical, open-source implementation that uses LDM for text-to-image generation.
 - ✓ Diffusion occurs in a 64×64 latent space for a 512×512 image.

Latent Diffusion Model Training Pipeline

latent-space equivalent of a
noisy image in DDPM.



Latent Diffusion Objective Function

- Instead of predicting noise in pixel space, the network predicts noise in latent space.

Training loss:

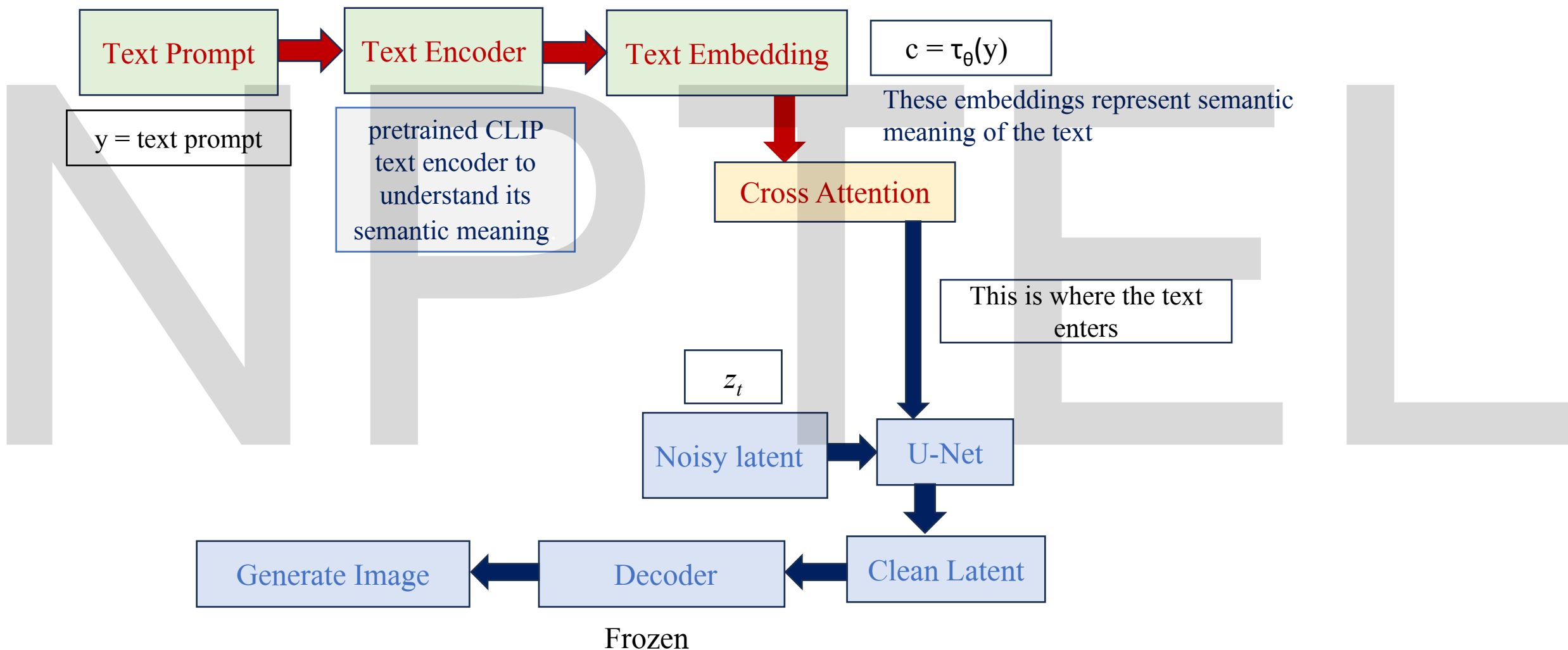
$$L = \mathbb{E}[\| \epsilon - \epsilon_{\theta}(z_t, t) \|^2]$$

where

- ✓ z_t = noisy latent
- ✓ ϵ = Gaussian noise
- ✓ ϵ_{θ} = predicted noise

This is the same DDPM objective, but applied to latent vectors

Stable Diffusion Pipeline (Text to Image Generation)



Cross Attention

Query

$$Q = W_Q^{(i)} \cdot \phi_i(z_t)$$

- ✓ Query is computed from intermediate U-Net feature maps.
- ✓ $W_Q^{(i)}$ learnable weight matrix
- ✓ $\phi_i(z_t)$: feature map produced by the i^{th} layer of the U-Net.
- ✓ Here, $\phi_i(z_t) \in \mathbb{R}^{N \times d}$ denotes a (flattened) intermediate representation of the U-Net

Key

$$K = W_K^{(i)} \tau_\theta(y)$$

- ✓ Key vector is computed from text embeddings:
- ✓ $W_K^{(i)}$: Learnable weight matrix
- ✓ $\tau_\theta(y)$: Text embeddings

Value

$$V = W_V^{(i)} \tau_\theta(y)$$

Value vector also comes from text embeddings

- $W_V^{(i)}$: Learnable weight matrix
- $\tau_\theta(y)$: Text embeddings

Cross Attention

The cross-attention weight :

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V$$

Why Cross attention is required:

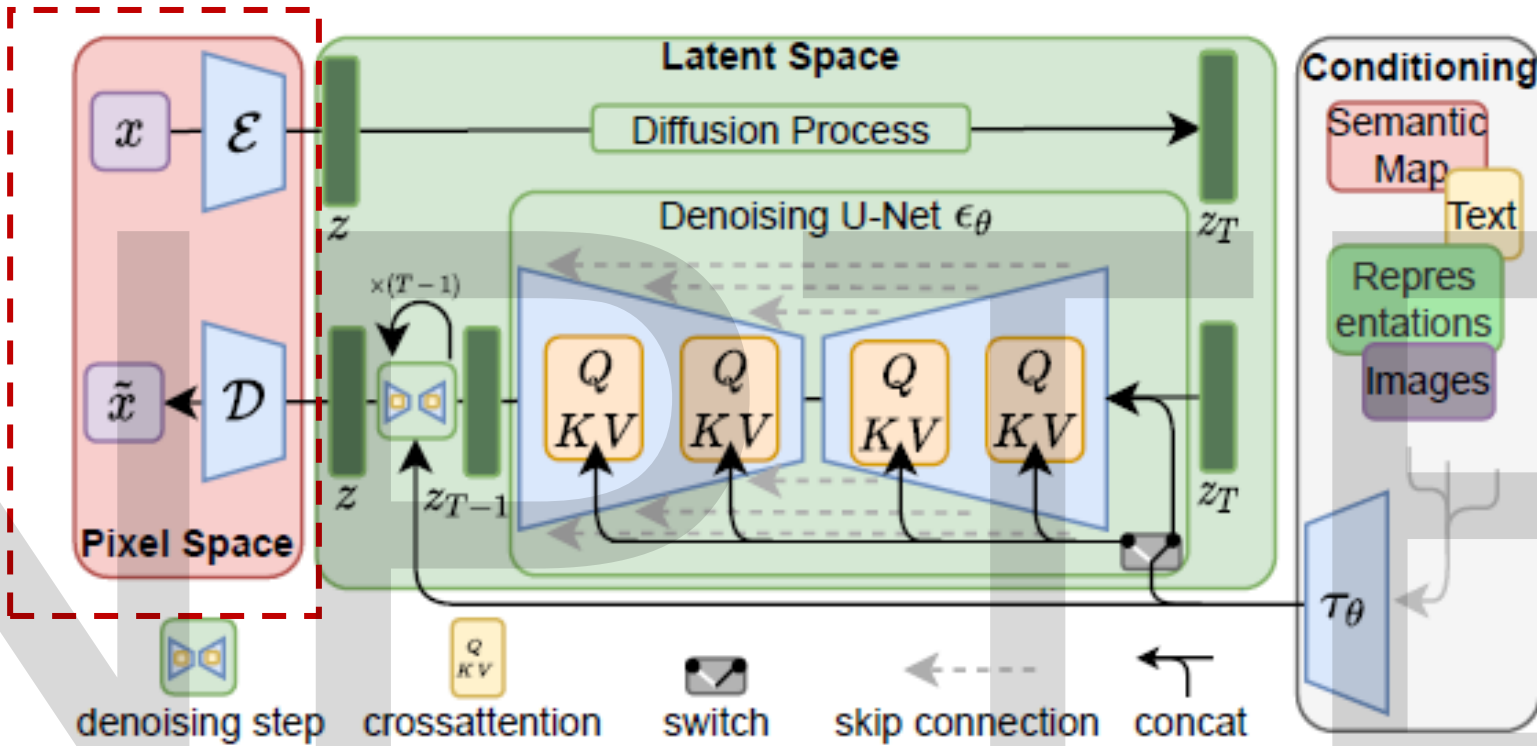
Example:

"A yellow butterfly sitting on a purple flower."

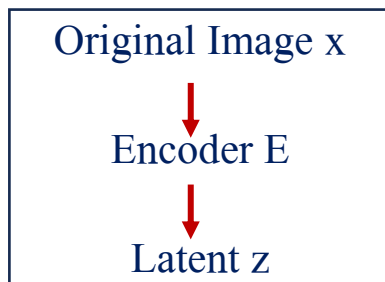
Different regions of the latent image need different words.

Latent region	Attends to
Butterfly	"yellow", "butterfly"
Flower	"purple", "flower"
Background	Remaining context

Architecture



Pixel Space

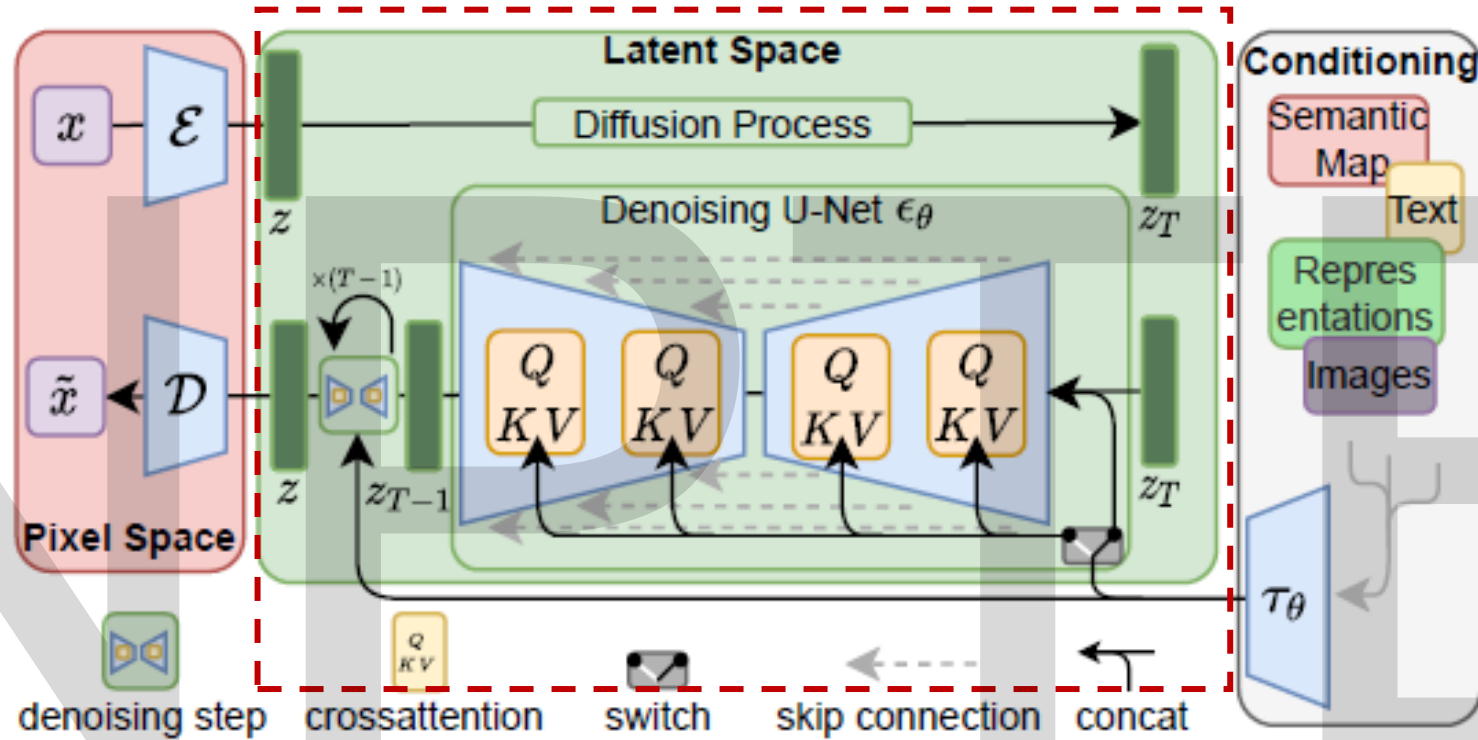


Decoder D reconstructs the image
 $\tilde{x} = D(z)$

Ref:

R. Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," Conference on Computer Vision and Pattern Recognition (CVPR) (2022)

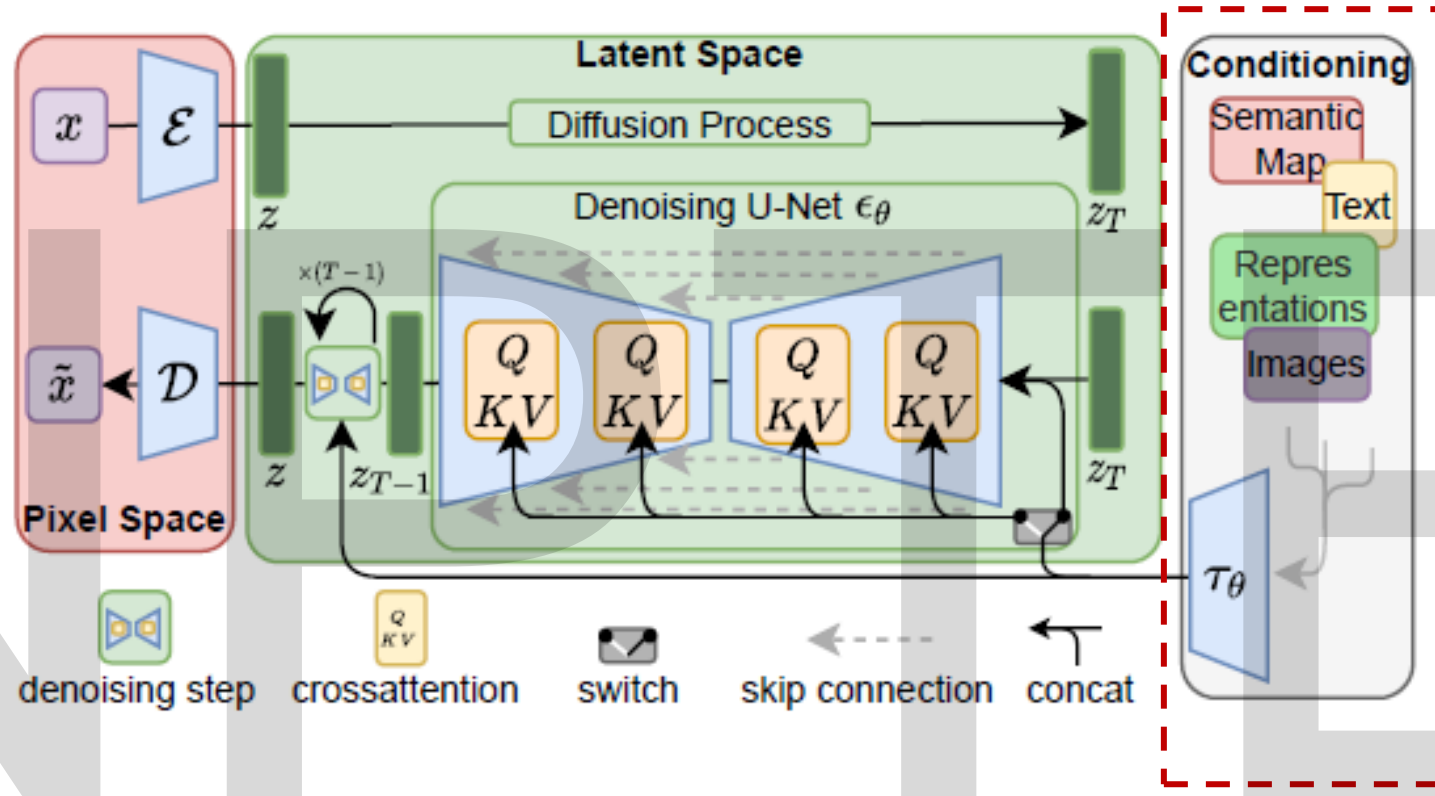
Architecture



Latent Space

- ✓ Diffusion occurs entirely in the latent space.
- ✓ Gaussian noise is progressively added to the latent representation z .
- ✓ The U-Net learns to remove this noise during reverse diffusion.
- ✓ Cross Attention is applied at several resolutions.

Architecture

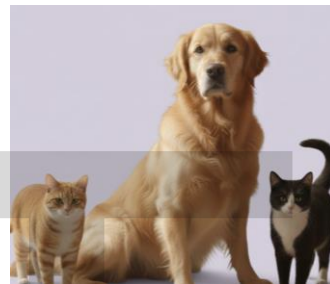


Stable Diffusion supports multiple conditioning modalities

- ✓ Text prompts
- ✓ Semantic maps
- ✓ Images

Conditioning

✓ Semantic Map to Image

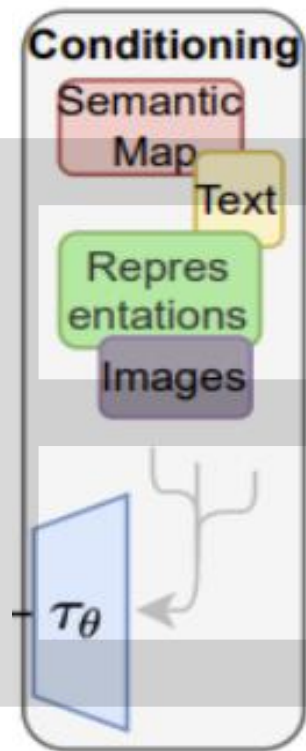


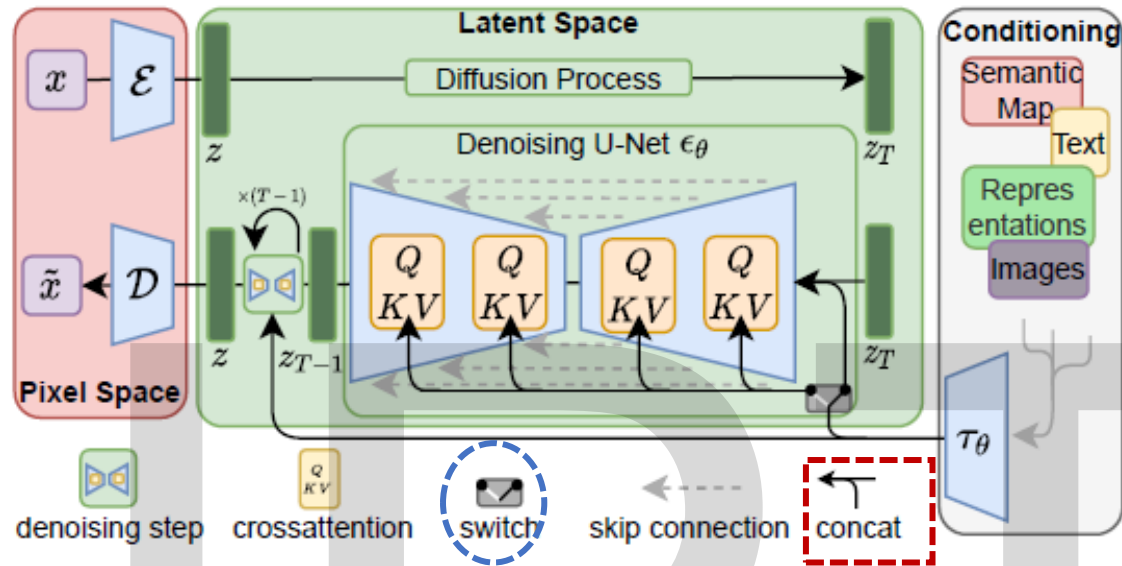
Text Prompting

The cat is sitting on a red carpet



Image to Image translation





Method 1:

Concatenation

- Used when conditioning has spatial information.
segmentation map
low-resolution image

These are concatenated with the latent feature maps.

Method 2

Cross Attention

- Used for Text embeddings (no spatial dimensions)

Switch:

- Spatial conditioning (segmentation maps, low-resolution images) uses concatenation.
- Text conditioning uses cross-attention

Advantage of Stable Diffusion over DDPM

DDPM

Adds Gaussian noise to a high-resolution photograph until it becomes pure noise.

Stable Diffusion

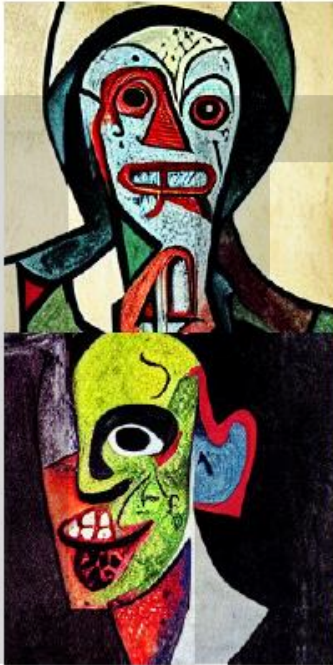
- First, compress the photograph into a compact "blueprint" (the latent),
- Then add Gaussian noise to that blueprint.
- During generation, the model cleans this noisy blueprint step by step and finally reconstructs the full image from it.
- This makes Stable Diffusion is much more computationally efficient than pixel-space diffusion.

Text-to-Image Synthesis on LAION. 1.45B Model.

'A street sign that reads
"Latent Diffusion" '



'A zombie in the
style of Picasso'



'An image of an animal
half mouse half octopus'



'An illustration of a slightly
conscious neural network'



'A painting of a
squirrel eating a burger'



'A watercolor painting of a
chair that looks like an octopus'



'A shirt with the inscription:
"I love generative models!" '



Ref:

R. Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," Conference on Computer Vision and Pattern Recognition (CVPR) (2022)

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Summary

NLP TREL

Next Session

Foundations of NLP